Course Title: SOFTWARE ENGINEERING

Semester: V

Academic Session: Aug-Dec 2026

Lab 6: Code and Branch Coverage Analysis

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Coverage_Analysis

1. Code Coverage

```
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6> python -m coverage run --source=airline_baggage -m
pytest -v CodeCoverageTest_PES2UG24CS652_application.py

CodeCoverageTest_PES2UG24CS652_application.py::test_invalid_cabin PASSED                                [  4%]
CodeCoverageTest_PES2UG24CS652_application.py::test_negative_weight PASSED                              [  8%]
CodeCoverageTest_PES2UG24CS652_application.py::test_member_economy_allowance PASSED                     [ 13%]
CodeCoverageTest_PES2UG24CS652_application.py::test_business_member_no_extra_allowance PASSED           [ 17%]
CodeCoverageTest_PES2UG24CS652_application.py::test_within_allowance PASSED                             [ 21%]
CodeCoverageTest_PES2UG24CS652_application.py::test_domestic_excess_fee PASSED                          [ 26%]
CodeCoverageTest_PES2UG24CS652_application.py::test_international_excess_fee PASSED                     [ 30%]
CodeCoverageTest_PES2UG24CS652_application.py::test_domestic_excess_above_20 PASSED                     [ 34%]
CodeCoverageTest_PES2UG24CS652_application.py::test_international_excess_above_20 PASSED                [ 39%]
CodeCoverageTest_PES2UG24CS652_application.py::test_infant_with_two_items PASSED                        [ 43%]
CodeCoverageTest_PES2UG24CS652_application.py::test_infant_with_more_than_two_items PASSED              [ 47%]
CodeCoverageTest_PES2UG24CS652_application.py::test_non_infant_within_weight_and_item_limit PASSED      [ 52%]
CodeCoverageTest_PES2UG24CS652_application.py::test_non_infant_over_weight_limit PASSED                 [ 56%]
CodeCoverageTest_PES2UG24CS652_application.py::test_non_infant_multiple_items PASSED                    [ 60%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_open_with_baggage PASSED                    [ 65%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_late_with_baggage PASSED                    [ 69%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_closed_with_baggage PASSED                  [ 73%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_open_without_baggage PASSED                 [ 78%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_late_without_baggage PASSED                 [ 82%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_closed_without_baggage PASSED               [ 86%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_open_international PASSED                   [ 91%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_late_international PASSED                   [ 95%]
CodeCoverageTest_PES2UG24CS652_application.py::test_checkin_closed_international PASSED                 [100%]

============================================== 23 passed in 0.17s ==============================================
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6> python -m coverage report -m
Name                Stmts   Miss  Cover   Missing
-------------------------------------------------
airline_baggage.py      37      0   100%
-------------------------------------------------
TOTAL                   37      0   100%
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6>
```

2. Branch Coverage

```
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6> python -m coverage erase
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6> python -m coverage run --branch --source=airline_bagga
ge -m pytest -v BranchCoverageTest_PES2UG24CS652_application.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.15, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\raksh\AppData\Local\Programs\Python\Python313\pytho
n.exe
cachedir: .pytest_cache
rootdir: C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6
collected 26 items

BranchCoverageTest_PES2UG24CS652_application.py::test_member_non_business PASSED                    [  3%]
BranchCoverageTest_PES2UG24CS652_application.py::test_non_member PASSED                              [  7%]
BranchCoverageTest_PES2UG24CS652_application.py::test_business_member PASSED                         [ 11%]
BranchCoverageTest_PES2UG24CS652_application.py::test_business_non_member PASSED                     [ 15%]
BranchCoverageTest_PES2UG24CS652_application.py::test_excess_exactly_20 PASSED                       [ 19%]
BranchCoverageTest_PES2UG24CS652_application.py::test_excess_greater_than_20_domestic PASSED         [ 23%]
BranchCoverageTest_PES2UG24CS652_application.py::test_excess_greater_than_20_international PASSED    [ 26%]
BranchCoverageTest_PES2UG24CS652_application.py::test_small_international_excess PASSED              [ 30%]
BranchCoverageTest_PES2UG24CS652_application.py::test_invalid_cabin PASSED                           [ 34%]
BranchCoverageTest_PES2UG24CS652_application.py::test_negative_weight PASSED                         [ 38%]
BranchCoverageTest_PES2UG24CS652_application.py::test_infant_allowed PASSED                          [ 42%]
BranchCoverageTest_PES2UG24CS652_application.py::test_infant_not_allowed PASSED                      [ 46%]
BranchCoverageTest_PES2UG24CS652_application.py::test_non_infant_weight_exactly_7 PASSED             [ 50%]
BranchCoverageTest_PES2UG24CS652_application.py::test_non_infant_weight_above_7 PASSED               [ 53%]
BranchCoverageTest_PES2UG24CS652_application.py::test_non_infant_items_exactly_one PASSED            [ 57%]
BranchCoverageTest_PES2UG24CS652_application.py::test_non_infant_items_above_one PASSED              [ 61%]
BranchCoverageTest_PES2UG24CS652_application.py::test_domestic_with_baggage_at_cutoff PASSED         [ 65%]
BranchCoverageTest_PES2UG24CS652_application.py::test_domestic_with_baggage_late_boundary PASSED     [ 69%]
BranchCoverageTest_PES2UG24CS652_application.py::test_domestic_with_baggage_closed PASSED            [ 73%]
BranchCoverageTest_PES2UG24CS652_application.py::test_domestic_without_baggage_at_cutoff PASSED      [ 76%]
BranchCoverageTest_PES2UG24CS652_application.py::test_domestic_without_baggage_late_boundary PASSED  [ 80%]
BranchCoverageTest_PES2UG24CS652_application.py::test_domestic_without_baggage_closed PASSED         [ 84%]
BranchCoverageTest_PES2UG24CS652_application.py::test_international_with_baggage PASSED              [ 88%]
BranchCoverageTest_PES2UG24CS652_application.py::test_international_without_baggage PASSED           [ 92%]
BranchCoverageTest_PES2UG24CS652_application.py::test_international_late PASSED                      [ 96%]
BranchCoverageTest_PES2UG24CS652_application.py::test_international_closed PASSED                    [100%]

============================================== 26 passed in 0.24s ==============================================
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6> python -m coverage report -m
Name                 Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------
airline_baggage.py      37      0     26      0   100%
----------------------------------------------------------------
TOTAL                   37      0     26      0   100%
PS C:\Users\raksh\Downloads\SE\new Lab\L6_L7_28_airline_baggage\lab6>
```